

**UNIVERSITY OF THE WITWATERSRAND
SCHOOL OF BUSINESS SCIENCES**

**INFORMATION SYSTEMS IIA (INFO2001A)
TEST 1 – AUGUST 2025_MEMO**

DURATION: 1¾ hours (105 minutes)

MARKS: 55

INSTRUCTIONS:

1. This paper consists of **EIGHT (8)** pages. Please ensure that you have them all.
 2. You must answer all questions and number them correctly.
 3. Sections A and B must be answered in the “**YELLOW**” booklet, while Section B must be answered in the “**GREEN**” booklet. Also put the Section(s) you are answering in the space for “**PAPER No.:**” on each booklet.
 4. You may draw diagrams in pencil.
 5. You may draw diagrams using landscape page orientation if necessary.
 6. You may **NOT** write in the right-hand margins – these are for marking purposes only.
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Section A

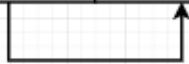
1. D
2. A
3. D
4. A
5. C
6. C
7. B
8. C
9. D
10. A

Section B

1. ENROLMENT (StudentID, StudentName, StudentEmail, CourseID, CourseName, SemesterID, SemesterStartDate, Grade, InstructorID, InstructorName)
 - {StudentID, CourseID, SemesterID} → {StudentName, StudentEmail, CourseName, SemesterStartDate, Grade, InstructorID, InstructorName} (all non-key attributes depend on the full key) (Full/Proper/Functional) ✓
 - StudentID → StudentName, StudentEmail (partial dependency) ✓
 - CourseID → CourseName (partial dependency) ✓
 - SemesterID → SemesterStartDate (partial dependency) ✓
 - InstructorID → InstructorName (non-key → non-key, giving a transitive path {StudentID, CourseID, SemesterID} → InstructorID → InstructorName) (Transitive) ✓
2. STUDENT (StudentID ✓, StudentName, StudentEmail) – Partial Dependency [3NF] ✓

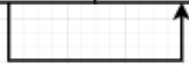
COURSE (CourseID ✓, CourseName) – Partial Dependency [3NF] ✓
SEMESTER (SemesterID ✓, SemesterStartDate) – Partial Dependency [3NF] ✓

<u>StudentID</u>	<u>CourseID</u>	<u>SemesterID</u>	StudentEmail	Grade	InstructorID	InstructorName
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3. INSTRUCTOR (InstructorID, StudentName) – Transitive Dependency [3NF]

<u>StudentID</u>	<u>CourseID</u>	<u>SemesterID</u>	StudentEmail	Grade	InstructorID	InstructorName
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The following table is in 3NF

<u>InstructorID</u>	InstructorName
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4. **Note:** According to the business rules, it was sufficient to resolve a single many-to-many relationship. For this reason, the cardinality between the **SEMESTER** class and the **COURSE** class is defined as **1..2**. However, if you did not assume that an academic year consists of exactly two semesters (and instead modeled the cardinality as **1..***), your solution would still be acceptable, but in that case, you would need to explicitly resolve the relationship.

